

Number Theory

Consider $m \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, m) = 1$.

Definition: For $(a, m) = 1$, $a \in \mathbb{Z}$, the order of a modulo m is the minimal $n \in \mathbb{N}$ s.t. $a^n \equiv 1 \pmod{m}$.

We write $n = \text{ord}_m a$.

Example: $\text{ord}_{100} 7 = 4$.

So with the order, we can generate the whole sequence of $a^n \pmod{m}$.

Since $a^{\varphi(m)} \equiv 1 \pmod{m}$, by Euler's Theorem, we can deduce that $\text{ord}_m a$ must divide $\varphi(m)$.

Thus $\text{ord}_m(a) \leq \varphi(m)$. What we focus today is when $\text{ord}_m a = \varphi(m)$.

Definition: a is a primitive root mod m if $\text{ord}_m a = \varphi(m)$.
In other words, the sequence $a, a^2, \dots, a^{\varphi(m)}$ form a reduced residue system mod m .

Example: 5 is a primitive root mod 7
2 " " " " " 11.

Example: (a) Let $(a, m) = 1$. If a has order $m-1$ mod m , show that m is prime.

(b) Let $n > 1$. Prove that $\varphi(2^n - 1)$ is divisible by n .

Solution:

(a) Note that $\varphi(m) \leq m-1$, by definition.
On the other hand, $(m-1) \mid \varphi(m)$.

$\therefore \varphi(m) = m-1$, m must be prime. $\square$

(b) Claim that $\text{ord}_{2^n-1} 2 = n$. Indeed,

For all $1 \leq k < n$, $2^k < 2^n - 1$. So as $2^n \equiv 1 \pmod{2^n-1}$,
we must have the claim. Since $\text{ord}_{2^n-1} 2 \mid \varphi(2^n-1)$,
 $n \mid \varphi(2^n-1)$. $\square$

Example: (a) Prove that 3 is a primitive root mod 17

(b) " " 2 " " " " 101.

(c) Find all primitive roots mod 29, between 1 & 28.

Solution: (a) By F.I.T., $3^{16} \equiv 1 \pmod{17}$. So $\text{ord}_{17} 3 \mid 16$.

So $\text{ord}_{17} 3$ can be 1, 2, 4, 8, 16. To show that $\text{ord}_{17} 3 = 16$,
it suffices to show that $3^8 \not\equiv 1 \pmod{17}$. Indeed, $3^8 = 6561$
 ~~$3^8 \equiv 1 \pmod{17}$~~ $\equiv -1 \pmod{17}$. $\therefore \text{ord}_{17} 3 = 16$, as desired. $\square$

(b) By F.I.T., $2^{100} \equiv 1 \pmod{101}$. So $\text{ord}_{101} 2 \mid 100$.

To show that $\text{ord}_{101} 2 = 100$, it suffices to show that

$2^{20} \not\equiv 1 \pmod{101}$ & $2^{50} \not\equiv 1 \pmod{101}$. Indeed, note that $2^{10} \equiv 14 \pmod{101}$,

so $2^{20} \equiv -6 \pmod{101}$, $2^{50} \equiv (-6)^2 (14) \pmod{101}$
 $\equiv -1 \pmod{101}$ as desired.

2 $\square$

(c) Claim that 2 is a primitive root mod 29.

In fact, for any ⁿ that is a primitive root mod 29, $m^{28} \equiv 1(29)$,

$$m^{14} \not\equiv 1(29) \text{ \& } m^4 \not\equiv 1(29).$$

$$\text{For } m=2, 2^4 \not\equiv 1(29) \text{ \& } 2^{14} = 16384 \equiv -1(29).$$

And $2^{28} = (16384)^2 \equiv 1(29)$. So claim follows.

Observe that all primitive roots mod 29 are of the form $2^k \pmod{29}$.

where $(k, 28) = 1$. $\therefore$ The primitive roots are

$$\begin{matrix} 1 & 3 & 5 & 9 & 11 & 13 & 15 & 17 & 19 \\ 2, & 8, & 3, & 19, & 18, & 14, & 27, & 21 & 26, \end{matrix}$$

$$\begin{matrix} 23 & 25 & 27 \\ 10 & 11 & 15. \end{matrix} \quad \text{Done.}$$

Example: Suppose that $n = 3^k$, $k \geq 1$. Prove that $3n \mid 2^n + 1$.

(Hint: 2 is a primitive root ~~of~~ ^{mod} 3^n , for any $n \geq 1$).

Proof: Since 2 is a primitive root mod 3^{k+1} , we know that

$$2^{4(3^{k+1})} \equiv 1 \pmod{3^{k+1}}, \text{ i.e. } 2^{2(3^k)} \equiv 1 \pmod{3^{k+1}}.$$

$$\text{Also, } 2^{3^k} \not\equiv 1 \pmod{3^{k+1}} \text{ \& } 2^{2(3^{k-1})} \not\equiv 1 \pmod{3^{k+1}}.$$

$$\therefore 2^{2(3^k)} - 1 \equiv 0 \pmod{3^{k+1}}.$$

$$(2^{3^k} + 1)(2^{3^k} - 1) \equiv 0 \pmod{3^{k+1}}.$$

$$2^{3^k} + 1 \equiv 0 \pmod{3^{k+1}} \text{ or } 2^{3^k} - 1 \equiv 0 \pmod{3^{k+1}} \quad (\text{rej.})$$

$$\therefore 3n \mid 2^n + 1.$$

□ 3

Solution: (a)

Example: (a) Let p be an odd prime. If q, r are primes s.t. $p \mid q^r + 1$, show that either $2r \mid p-1$ or $p \mid q^2 - 1$.

(b) Let $a, n \in \mathbb{N}$, $a > 1$. If prime $p \mid a^{2^n} + 1$, show that $2^{n+1} \mid p-1$.

Proof: (a) $p \mid q^r + 1 \iff q^r \equiv -1 \pmod{p}$, so $q^{2r} \equiv 1 \pmod{p}$.

Let $d = \text{ord}_p q$. Then $d \mid 2r$ & $d \neq r$. Since r is prime,

$\therefore d = 2$ or $2r$ (note that d cannot be 1) ~~as p odd, $q \not\equiv \pm 1 \pmod{p}$~~

If $d=2$, then $q^2 \equiv 1 \pmod{p}$, $p \mid q^2 - 1$. ✓ Yay ☺

If $d=2r$, then since $d \mid \varphi(p)$, $2r \mid p-1$. ✓ □

(b) $p \mid a^{2^n} + 1 \iff a^{2^n} \equiv -1 \pmod{p}$, so $a^{2^{n+1}} \equiv 1 \pmod{p}$.

Let $d = \text{ord}_p a$. Then $d \mid 2^{n+1}$, d must be a power of 2.

Since $a^{2^n} \equiv -1 \pmod{p}$, ~~$d \neq 2^n$~~ $d \neq 2^n$. Thus $d = 2^{n+1}$.

$\therefore d \mid \varphi(p)$, $\therefore 2^{n+1} \mid p-1$. □

Example: (a) Find all nonnegative integers x, y s.t. $1 + 3^x = 2^y$.

(b) " " " " " " $1 + 2^x = 3^y$

(c) " " " " " " x, y, z " $3^x + 4^y = 5^z$.

Solution: (a) Since $\text{ord}_8 3 = 2$, $3^x \equiv 1, 3 \pmod{8}$,

$$1 + 3^x \equiv 2, 4 \pmod{8}. \therefore y \leq 2, \text{ i.e. } y = 1, 2.$$

For $y=1$, $x=0$; for $y=2$, $x=1$.

$$\therefore (x, y) = (0, 1), (1, 2).$$

(b) Since $\text{ord}_{16} 3 = 4$, $3^y \equiv 1, 3, 9, 11 \pmod{16}$.

$$3^y - 1 \equiv 0, 2, 8, 10 \pmod{16}.$$

If $x \geq 4$, $2^x \equiv 0 \pmod{16}$. Then $y \equiv 0 \pmod{4}$.

Then write $y = 4k$, $k \in \mathbb{Z}_{\geq 0}$. The equation becomes

$$1 + 2^x = 81^k.$$

$\text{mod } 10$ gives $1 + 2^x \equiv 1 \pmod{10}$, $2^x \equiv 0 \pmod{10}$. ϕ .

So $x = 0, 1, 2, 3$.

$$\therefore (x, y) = (1, 1) \text{ \& } (3, 2).$$

(c) For $x \neq 0$, $\text{mod } 4$ gives $(-1)^x \equiv 1^z \pmod{4}$. $\therefore x$ even.

Let $x = 2x_1$, $x_1 \in \mathbb{N}$. Then equation becomes $9^{x_1} + 4^y = 5^z$.

for $x=0$, $y=1, z=1$. ~~$\text{mod } 5$ gives $(-1)^{x_1} + (-1)^y \equiv 0 \pmod{5}$.~~

$\text{mod } 3$ gives $1 \equiv (-1)^z \pmod{3}$. $\therefore z$ even.

Let $z = 2z_1$, $z_1 \in \mathbb{N}$. Then equation becomes $9^{x_1} + 4^y = 25^{z_1}$.

$$9^{x_1} = 25^{z_1} - 4^y$$

$$9^{x_1} = (5^{z_1} - 2^y)(5^{z_1} + 2^y)$$

Since $5^{z_1} + 2^y$ & $5^{z_1} - 2^y$ cannot have factor of 3

together, we have
$$\begin{cases} 5^{z_1} + 2^y = 9^{x_1} \\ 5^{z_1} - 2^y = 1 \end{cases}$$

$$\text{mod } 3 \Rightarrow \begin{cases} (-1)^{z_1} + (-1)^y \equiv 0 \pmod{3} \\ (-1)^{z_1} - (-1)^y \equiv 1 \pmod{3} \end{cases}$$

adding, $2(-1)^{z_1} \equiv 1 \pmod{3}$.

$$(-1)^{z_1} \equiv -1 \pmod{3}. \quad \therefore z_1 \text{ odd.}$$

mod 8 $\Rightarrow$ for $y \geq 3$, $5^{z_1} \equiv 1 \pmod{8}$, z_1 can only be even. ~~✗~~

$$\therefore y < 3.$$

By trial & error, $y=2, z_1=1, z=2, x=2$
 $y=1,$
 $\therefore (x, y, z) = (2, 2, 2).$ ☺